12

Black Holes Evaporate

in which a black-hole horizon
is clothed in an atmosphere
of radiation and hot particles
that slowly evaporate,
and the hole shrinks
and then explodes

Black Holes Grow

The Idea hit Stephen Hawking one evening in November 1970, as he was preparing for bed. It hit with such force that he was left almost gasping for air. Never before or since has an idea come to him so quickly.

Preparing for bed was not easy. Hawking's body is afflicted with amyotrophic lateral sclerosis (ALS), a disease that gradually destroys the nerves which control the body's muscles and leaves the muscles, one after another, to waste away in disuse. He moved slowly, with legs wobbling and at least one hand always firmly grasping a countertop or bedpost, as he brushed his teeth, disrobed, struggled into his pajamas, and climbed into bed. That evening he moved even more slowly than usual, since his mind was preoccupied with the Idea. The Idea excited him. He was ecstatic, but he didn't tell his wife, Jane; that would have made him most unpopular, since he was supposed to be concentrating on getting to bed.

He lay awake for many hours that night. He couldn't sleep. His mind kept roaming over the Idea's ramifications, its connections to other things.

The Idea had been triggered by a simple question. How much gravitational radiation (ripples of spacetime curvature) can two black holes